Regularization and Stability

Data Science for Economics - Statistical Methods for Machine Learning

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1 Introduction

A crucial question in Machine Learning discusses about why an algorithm that fits its training data well also performs well on new data. The classical answer relies on the complexity of the hypothesis class. Bousquet and Elisseeff [1] propose an alternative viewpoint based on *algorithmic stability*: instead of describing the class of predictors, they look at the learning algorithm itself and evaluate how much its output changes when a single training example is removed or replaced. Algorithms whose output is only weakly affected by such a change are considered *stable*, and it can be proved that stability controls the generalisation error.Regularisation is a natural source of stability, because the penalty prevents the solution from depending too strongly on any single example.

Goal of this project. We study this viewpoint empirically for least squares by implementing ordinary least squares (OLS) and ridge regression from scratch. On a real and a synthetic regression dataset, we (i) measure the training and test error as a function of the regularisation strength λ; (ii) measure stability directly, by removing one training example, retraining, and recording the change on a fixed test set, and see how it varies with λ; (iii) relate the measured stability to the generalisation gap; (iv) as an extension, study how stability varies with the training set size m.

Section 2 presents the algorithms, their implementation and the stability measures, Section 3 the datasets, Section 4 the experimental protocol, Section 5 the results, Section 6 the extension on m, and Sections 7–8 give the discussion and conclusion.

2 Methods

2.1 Setting

We consider regression with a training set $S = \{(x_i, y_i)\}_{i=1}^m$, $x_i \in \mathbb{R}^d$, $y_i \in \mathbb{R}$, and a linear predictor $f(x) = w^\top x + b$, where $w \in \mathbb{R}^d$ are the weights and $b \in \mathbb{R}$ is the intercept, a constant offset that, unlike w, is not penalised by the regularisation (Section 2.2). The loss is the square loss $\ell(f, (x, y)) = (f(x) - y)^2$. Errors are reported as mean squared errors (MSE) on the training set and on a separate test set that is never used for fitting.

2.2 Ordinary least squares and ridge regression

Let $X \in \mathbb{R}^{m \times d}$ be the matrix with rows $x_i^\top$ and $y \in \mathbb{R}^m$ the vector of targets.

Ridge regression. For $\lambda > 0$, ridge regression minimises over $(w, b) \in \mathbb{R}^d \times \mathbb{R}$

$$\frac{1}{m} \sum_{i=1}^{m} \big(w^\top x_i + b - y_i\big)^2 + \frac{\lambda}{2} \|w\|^2. \tag{1}$$

The first term is the training error and the second shrinks the weights; λ sets the trade-off between the two. Only w is penalised, not the intercept b.

Closed form. Let $\bar{x} = \frac{1}{m} \sum_i x_i$ and $\bar{y} = \frac{1}{m} \sum_i y_i$ be the training means, and let X_c and y_c be X and y with $\bar{x}^\top$ and $\bar{y}$ subtracted from every row and entry (centring). Minimising (1) over b gives $b = \bar{y} - \bar{x}^\top w$. Substituting this back leaves a ridge problem for w alone, $\frac{1}{m} \|X_c w - y_c\|^2 + \frac{\lambda}{2} \|w\|^2$, whose gradient vanishes at

$$w = \big(X_c^\top X_c + \alpha I\big)^{-1} X_c^\top y_c, \qquad b = \bar{y} - \bar{x}^\top w, \qquad \alpha = \frac{m\lambda}{2}. \tag{2}$$

For $\lambda > 0$ the matrix $X_c^\top X_c + \alpha I$ is positive definite, hence invertible, and the objective is strictly convex, so (2) is the unique minimiser. The parameter α is the penalty of the equivalent sum form $\|X_c w - y_c\|^2 + \alpha\|w\|^2$: multiplying (1) by m turns the average of the squared errors into their sum and the penalty into $\frac{m\lambda}{2}\|w\|^2$, without changing the minimiser.

Ordinary least squares. Ordinary least squares (OLS) is the unpenalised case $\lambda = 0$. If $X_c^\top X_c$ is invertible and well conditioned (condition number below 10^{10}), we solve the normal equations $X_c^\top X_c\, w = X_c^\top y_c$. Otherwise (e.g. collinear features or $d \geq m$) the minimiser is not unique or not numerically reliable, and we return the minimum-norm least-squares solution $w = X_c^+ y_c$ computed from the singular value decomposition (SVD). In both cases $b = \bar{y} - \bar{x}^\top w$.

Implementation. All algorithms are implemented using NumPy linear algebra routines; scikit-learn is used only to load the Diabetes data and Matplotlib only for the figures.

2.3 Measuring stability

Let f_S be the model trained on S and $f_{S\setminus i}$ the model trained on S with the i-th example removed. To compare the two models under the same regularised problem, the penalty $\alpha = m\lambda/2$ of (2) is kept at its value for the full sample, i.e. it is computed with m and not with $m-1$, so that the two models differ only in the data. Stability is estimated on a fixed test set $\{x_j\}_{j=1}^n$ with two measures, both averaged over the m possible removals.

Prediction stability. The first measure is the average change of the predictions on the test set,

$$\hat{\beta}_{\text{pred}} = \frac{1}{m}\sum_{i=1}^{m}\frac{1}{n}\sum_{j=1}^{n}\big|\, f_S(x_j) - f_{S\setminus i}(x_j)\,\big|. \tag{3}$$

Because the intercept is not penalised, it still changes when an example is removed, however large λ is: as $\lambda \to \infty$, $w \to 0$ and $b = \bar{y}$, so $\hat{\beta}_{\text{pred}}$ tends to a positive floor of about $\overline{|y_i - \bar{y}|}/(m-1)$, the average change of $\bar{y}$ when one example is removed. Since regularisation acts only on w, our main curves use the weights-only version, in which the predictions are replaced by $(x_j - \bar{x})^\top w$:

$$\hat{\beta}^{w}_{\text{pred}} = \frac{1}{m}\sum_{i=1}^{m}\frac{1}{n}\sum_{j=1}^{n}\big|\,(x_j - \bar{x})^\top (w_S - w_{S\setminus i})\,\big|.$$

Loss stability. The second measure is the average change of the loss on the removed example itself,

$$\hat{\beta}_{\text{loss}} = \frac{1}{m}\sum_{i=1}^{m}\Big[\ell\big(f_{S\setminus i}, z_i\big) - \ell\big(f_S, z_i\big)\Big]. \tag{4}$$

This quantity equals the leave-one-out error minus the training error, so it estimates the generalisation gap. For both measures we also record the maximum over i in place of the average, as a worst-case counterpart.

2.4 Correctness checks

Self-checks are done so that the closed-form solution satisfies the optimality condition of (1), and that the leave-one-out loss gaps computed by retraining agree with those obtained exactly in closed form from the hat matrix.

	Diabetes (real)	Synthetic
Features d	10	10
Training size m	100	100
Test size	342	5000
Sample	442 patients (fixed)	new draw per seed
Condition number of feature correlations	470	420

Table 1: The two datasets in the main experiments.

3 Datasets

We use one real and one synthetic regression dataset, summarised in Table 1.

Diabetes (real-world data). The Diabetes dataset contains 442 patients, with ten baseline measurements (age, sex, body mass index, blood pressure and six blood serum measurements) and a quantitative measure of disease progression one year later as the target. It is loaded with scikit-learn in its raw, unscaled form. For each random seed, 100 patients are drawn as the training set and the remaining 342 form the test set. The features are strongly correlated: the condition number of their correlation matrix is about 470, so OLS is expected to be sensitive to the training sample.

Synthetic data. The synthetic problem has known data-generating distribution: $x \sim \mathcal{N}(0, \Sigma)$ with $\Sigma_{ij} = \rho^{|i-j|}$, $d = 10$, and $y = w_\star^\top x + \varepsilon$ with $\varepsilon \sim \mathcal{N}(0, \sigma^2)$. The true weights $w_\star$ are drawn once from $\mathcal{N}(0, I)$ with a fixed seed and then kept fixed. We set $\rho = 0.96$ so that the condition number of Σ (about 420) is comparable to that of Diabetes, and $\sigma^2 = w_\star^\top \Sigma w_\star$, so that the population R^2 equals 0.5. For each seed a new training set of 100 points and a new test set of 5000 points are drawn. The synthetic data serve two purposes: the true noise level gives a reference value for the test error (the Bayes risk), and the difficulty of the problem (correlation, noise, sample size) can be controlled, which allows a comparison with the real data under similar conditioning.

Training size. The default training size is $m = 100$, that is $m/d = 10$. It is chosen small enough for the variance of the fitted model to matter: on Diabetes, the test $R^2 = 1 - \mathrm{MSE}/\mathrm{Var}(y_{\mathrm{test}})$ of OLS (average over 50 random splits) is 0.39 for $m = 60$ and 0.45 for $m = 100$, but about 0.47 for $m = 200$ and $m = 353$, where there is little left for regularisation to improve. This value is a design choice and not an optimum; Section 6 varies m and checks how the conclusions depend on it.

Preprocessing and absence of data leakage. The split into training and test sets is made before any preprocessing. The features are standardised with the mean and standard deviation of the *training* set only, and the same transformation is then applied to the test set. The targets are divided by the standard deviation of the *training* targets, so that every error is expressed as a fraction of the training label variance and the values of λ are comparable across the two datasets. The test set is used only to evaluate the models; the regularisation strength is never chosen on it (Section 4.4).

4 Experiments

4.1 Common protocol

All experiments use the training size $m = 100$ and are repeated for 50 random seeds (0–49). For each seed, a new training/test split is made: on Diabetes, 100 of the 442 patients are drawn at random for training and the others form the test set; on the synthetic data, a new sample is generated. The same split is used for all values of λ, so that comparisons across λ are paired. The regularisation strength is taken on a grid of 30 values, logarithmically spaced between 10^{-4} and 10^{3}, together with $\lambda = 0$ (OLS) as a reference. Every reported quantity is the average over the 50 seeds, and the shaded bands in the figures show the standard error of the mean (standard deviation divided by $\sqrt{50}$).

4.2 Training and test error

For every seed and every λ we fit ridge regression on the training set and record its training and test MSE, together with those of OLS. Errors are expressed as a fraction of the training label variance. The result is the training and test error of ridge regression as a function of λ, with OLS as the baseline.

4.3 Stability and its relation to generalisation

For every seed and every λ we apply the leave-one-out protocol: each of the 100 training examples is removed in turn, the model is retrained with the penalty α kept fixed, and the changes on the test set and on the removed example are recorded. This gives the weights-only prediction stability $\hat{\beta}^{w}_{\text{pred}}$ (main curve), the total prediction stability $\hat{\beta}_{\text{pred}}$ and the loss stability $\hat{\beta}_{\text{loss}}$, each averaged over the removals. We then compare:

- the dependence of $\hat{\beta}^{w}_{\text{pred}}$ on λ, and its decay in the large-λ regime ($\lambda \geq 10$), measured as the slope of $\log \hat{\beta}^{w}_{\text{pred}}$ against $\log \lambda$ and compared with the slope -1 that corresponds to a $1/\lambda$ decay;
- $\hat{\beta}_{\text{loss}}$ with the generalisation gap, defined as the test MSE minus the training MSE, across the values of λ.

4.4 Selecting λ

To report a test error for ridge regression without choosing λ on the test set, λ is selected by 10-fold cross-validation on the training set only, over the same grid of 30 positive values; the folds are a random partition of the training examples. The model is then refitted on the whole training set with the selected λ and evaluated once on the test set. Its test error is compared with that of OLS through the paired difference over the 50 seeds. For reference only, we also report the test error obtained with the λ that minimises the test error over the grid. This value uses the test set, so it is optimistic and is never used as a selection method.

5 Results

5.1 Training and test error

Figure 1 shows the training and test error as a function of λ, and Table 2 summarises the key values. With OLS, the training error is 0.451 (Diabetes) and 0.442 (synthetic), while the test error is 0.565 and 0.572. On the synthetic data, the test error of OLS lies above the Bayes risk of 0.511, which is the smallest error any predictor can reach.

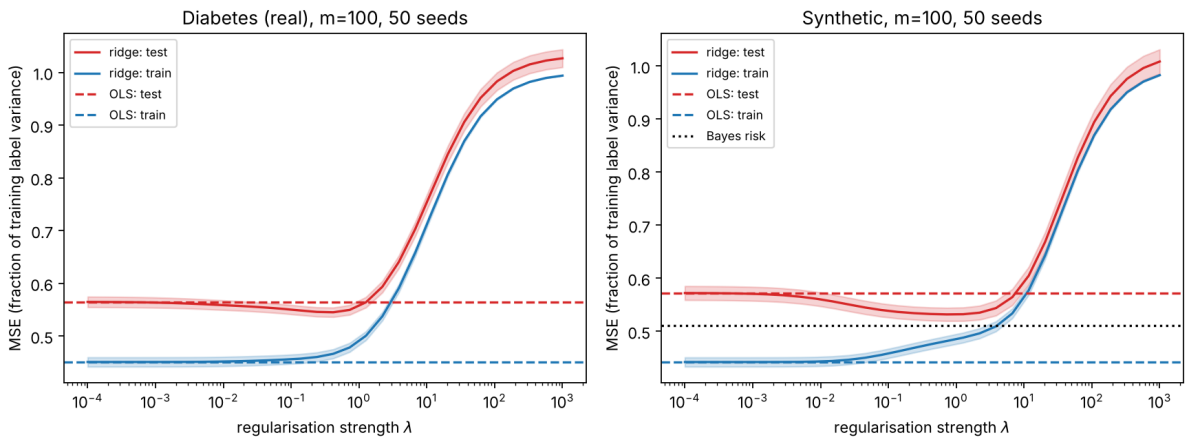


Figure 1: Training and test MSE of ridge regression against λ (mean over 50 seeds, bands: standard error). Dashed lines: OLS. Dotted line: Bayes risk of the synthetic data.

	Diabetes	Synthetic
OLS: train / test	0.451 / 0.565	0.442 / 0.572
Ridge at λ^*: train / test	0.466 / 0.545	0.482 / 0.532
λ^* (minimum of the mean test curve)	0.42	0.73
Bayes risk	–	0.511

Table 2: Training and test MSE (fraction of training label variance). λ^* is the value on the grid that minimises the mean test error; it uses the test set and is only descriptive.

As λ increases, the training error of ridge regression increases monotonically. The test error first decreases slightly below that of OLS, reaches a minimum at $\lambda^* \approx 0.42$ (Diabetes, test error 0.545, 3.5% below OLS) and $\lambda^* \approx 0.73$ (synthetic, 0.532, 7.1% below OLS), and then increases. For $\lambda \geq 10$ both errors rise quickly towards 1, the error of a model that predicts the training mean. The gain over OLS is small on both datasets and, since λ^* is chosen on the test set, it is optimistic; a legitimate estimate is given in Section 5.4.

5.2 Stability as a function of λ

Figure 2 shows the prediction stability, where lower values mean a more stable algorithm. For OLS, removing one training example changes the test predictions by 0.0156 (Diabetes) and 0.0158 (synthetic) on average (weights only). The weights-only stability decreases monotonically as λ increases: it is 0.0090 and 0.0050 at $\lambda = 1$ and 0.00060 and 0.00092 at $\lambda = 100$ (values interpolated on the grid), more than an order of magnitude below OLS. At the value of λ that minimises the mean test error, it is already lower than for OLS (0.0109 against 0.0156 on Diabetes, 0.0054 against 0.0158 on the synthetic data).

For large λ the decay is close to, but slightly slower than, $1/\lambda$. The slope of $\log \hat{\beta}^{w}_{\text{pred}}$ against $\log \lambda$ for $\lambda \geq 10$ is -0.89 on Diabetes and -0.73 on the synthetic data, to be compared with -1. On the synthetic data the curve is flatter at intermediate λ and steepens towards the largest values. The total stability, which includes the intercept, decreases at first but then levels off at a positive value instead of going to zero, as explained in Section **??**. The worst case over the removed examples is larger than the average: for OLS the maximum change is about 4.5 times (Diabetes) and 3.9 times (synthetic) the average.

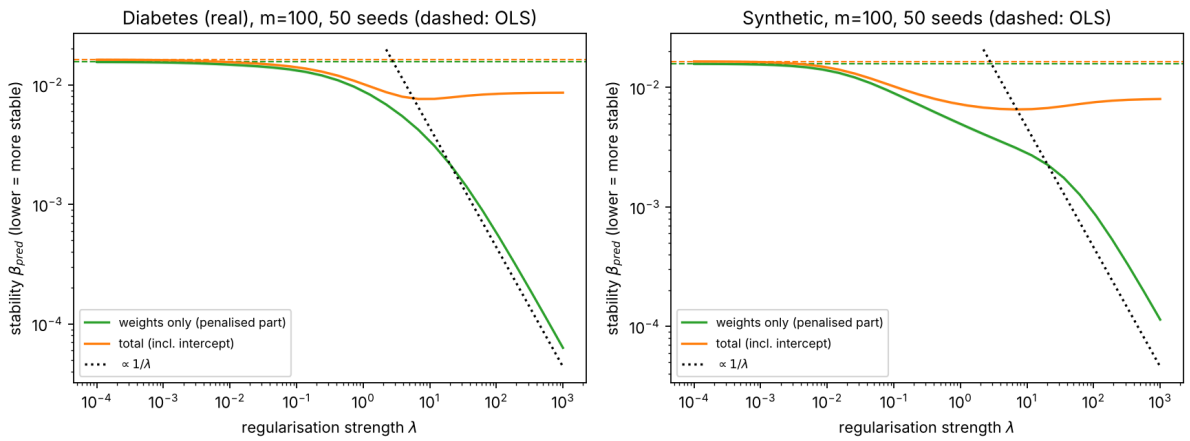


Figure 2: Prediction stability against λ on logarithmic axes (mean over 50 seeds, bands: standard error). Green: weights only; orange: total, including the intercept. Dashed: OLS. Dotted: a $1/\lambda$ line for reference.

5.3 Stability and generalisation

Figure 3 plots the loss stability $\hat{\beta}_{\text{loss}}$ against the generalisation gap, with one point per value of λ. Both decrease as λ increases, and the two quantities are almost perfectly correlated across λ (correlation 0.999 on Diabetes and 0.998 on the synthetic data). For OLS, $\hat{\beta}_{\text{loss}} = 0.119$ against a gap of 0.114 on Diabetes, and 0.117 against 0.130 on the synthetic data. On Diabetes the points lie close to the line $y = x$, so $\hat{\beta}_{\text{loss}}$ is a good estimate of the gap; on the synthetic data it lies below the gap for intermediate λ, that is it underestimates it.

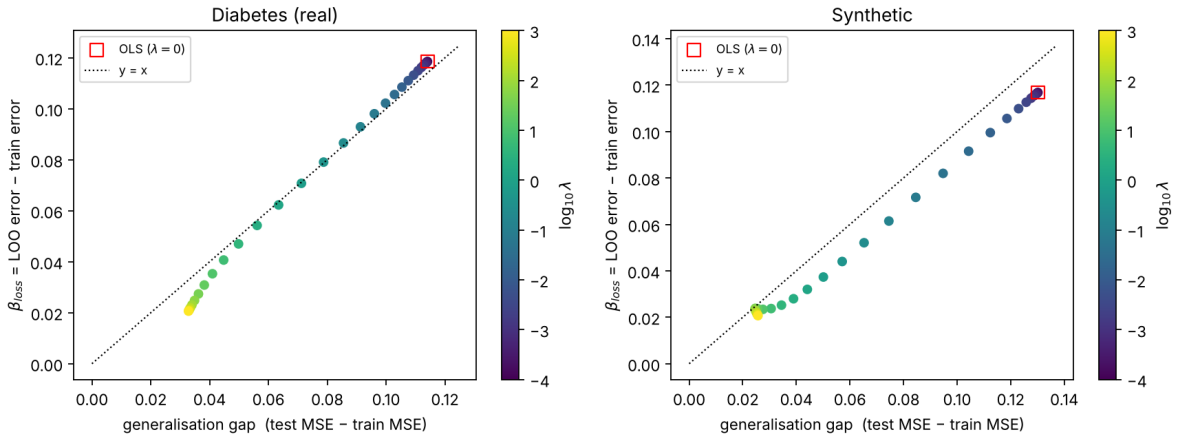


Figure 3: Loss stability $\hat{\beta}_{\text{loss}}$ (leave-one-out error minus training error) against the generalisation gap (test MSE minus training MSE). Each point is a value of λ (colour), averaged over 50 seeds; the red square is OLS.

5.4 Choosing λ by cross-validation

Table 3 compares OLS with ridge regression when λ is selected by 10-fold cross-validation on the training set. The median selected values are 0.42 (Diabetes) and 0.73 (synthetic). On the synthetic data ridge regression improves on OLS in all 50 seeds, with a paired gain of 0.037 ± 0.003 (mean $\pm$ standard error). On Diabetes the gain is smaller, 0.011 ± 0.004, and ridge regression is better in 39 of the 50 seeds. The per-seed oracle, which picks the best λ on the test set for every

seed, gains more on both datasets (0.0235 against 0.0113 on Diabetes, 0.0438 against 0.0371 on the synthetic data), which shows how optimistic it is. With the selected λ the algorithm is also more stable than OLS.

	Diabetes	Synthetic
OLS, test MSE	0.5650	0.5723
Ridge, λ by cross-validation, test MSE	0.5537	0.5352
paired gain over OLS ($\pm$ s.e.)	0.0113 ± 0.0039	0.0371 ± 0.0031
seeds where ridge is better	39/50	50/50
Ridge, per-seed oracle λ, test MSE	0.5415	0.5285
paired gain over OLS ($\pm$ s.e.)	0.0235 ± 0.0045	0.0438 ± 0.0040
Stability (weights only): OLS $\to$ ridge with CV λ	$0.0156 \to 0.0113$	$0.0158 \to 0.0060$

Table 3: Test MSE of OLS and of ridge regression with λ chosen by cross-validation (50 seeds). The oracle selects λ separately for each seed on the test set and is shown only as a reference.

6 Extension: Effect of the Training Set Size

The theory predicts that stability also improves with the sample size m, at a rate of order $1/m$ for a fixed λ. To test this, we vary $m \in \{30, 50, 75, 100, 150, 200, 300\}$ for $\lambda \in \{0 \text{ (OLS)}, 0.01, 0.1, 1, 10\}$, with 30 seeds for each combination, and measure the weights-only prediction stability with the same removal protocol as before. Since λ is held fixed, the penalty $\alpha = m\lambda/2$ grows with m. To make the values comparable across m, the Diabetes test set is fixed to 100 patients that are drawn first, and the training set is taken from the other patients; the synthetic data keep their 5000 test points.

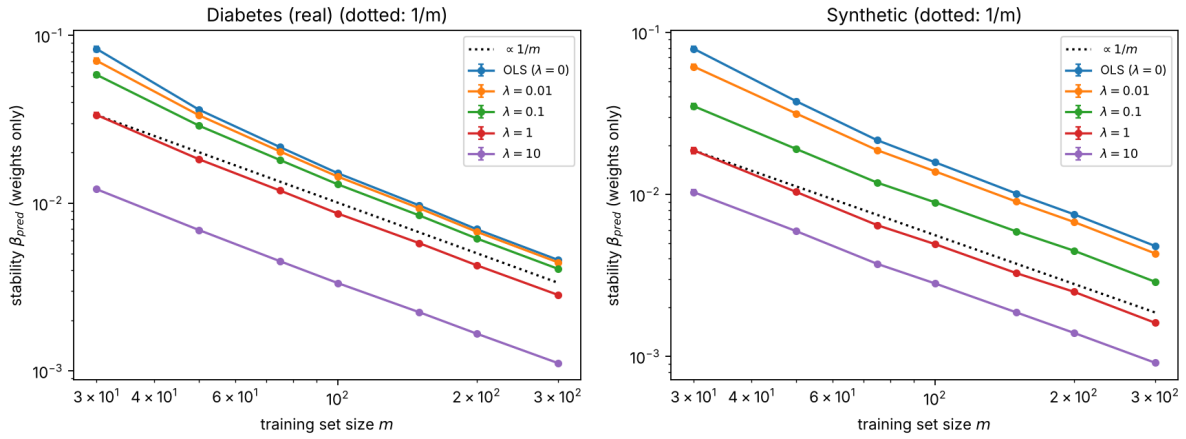


Figure 4: Weights-only prediction stability against the training size m on logarithmic axes (mean over 30 seeds, bars: standard error). Dotted: a $1/m$ line anchored at $\lambda = 1$, $m = 30$.

Figure 4 shows that, for every λ, the stability improves as m grows, and that at every m a larger λ gives a more stable algorithm. On log-log axes the curves are close to straight lines parallel to the $1/m$ reference. Table 4 gives the slopes fitted for $m \geq 50$: they lie between -1.02 and -1.15, close to the value -1 that corresponds to a $1/m$ decay. The slope is steeper than -1 for OLS and small λ and approaches -1 as λ increases. For OLS on Diabetes, the stability changes from 0.083 at $m = 30$ to 0.0046 at $m = 300$, and with $\lambda = 10$ from 0.0121 to 0.0011. The generalisation gap (test minus training MSE) of OLS also shrinks with m, from 0.543 to

0.047 on Diabetes and from 0.538 to 0.034 on the synthetic data; for $\lambda = 10$ and $m = 300$ it is 0.023 and -0.005, respectively.

λ	0 (OLS)	0.01	0.1	1	10
Diabetes	-1.15	-1.12	-1.09	-1.04	-1.02
Synthetic	-1.13	-1.09	-1.04	-1.02	-1.03

Table 4: Slope of $\log \hat{\beta}^{w}_{\text{pred}}$ against $\log m$, fitted on $m \in \{50, \ldots, 300\}$.

7 Discussion

Findings. On both datasets, regularisation makes the algorithm more stable (Figure 2), stability improves with the training size at a rate close to $1/m$ (Table 4), and the loss stability follows the generalisation gap (Figure 3). The price is bias: the test error is lowest at a moderate λ and rises towards that of the mean predictor for large λ (Figure 1). The gain over OLS is modest: with λ chosen by cross-validation it is 0.011 ± 0.004 on Diabetes and 0.037 ± 0.003 on the synthetic data (Table 3).

Relation to the theory. The measured decay (slopes -0.89 and -0.73 in λ for $\lambda \geq 10$, and between -1.02 and -1.15 in m) agrees in trend with $1/\lambda$ and $1/m$, but we do not verify the bounds themselves: they assume a bounded or Lipschitz loss and a supremum over all points, whereas we use the square loss with unbounded data and average over a finite test set.

Limitations. (i) The unpenalised intercept prevents the total prediction stability from going to zero, so the main curve uses the weights only, which is our own choice. (ii) The correlation between $\hat{\beta}_{\text{loss}}$ and the gap across λ is partly due to both decreasing with λ, and on the synthetic data $\hat{\beta}_{\text{loss}}$ underestimates the gap. (iii) We only remove examples, we do not replace them, and we use two datasets with $d = 10$ and a default $m = 100$; the synthetic data resemble Diabetes only in conditioning and noise level. (iv) The λ^* of Table 2 and the oracle of Table 3 use the test set and are optimistic.

8 Conclusion

We implemented OLS and ridge regression from scratch and measured, on a real and a synthetic dataset, the training and test error and the stability of the algorithm when one training example is removed. Regularisation increases stability monotonically, with a decay for large λ that is close to, though slightly flatter than, $1/\lambda$, at the price of a higher bias; the test error improves only modestly and only for moderate λ. Stability also improves with the training size at a rate close to $1/m$, and the loss stability follows the generalisation gap. These results agree in trend with the stability viewpoint of Bousquet and Elisseeff, although the bounds themselves are not tested.

References

[1] O. Bousquet and A. Elisseeff. Stability and generalization. *Journal of Machine Learning Research*, 2:499–526, 2002.